
SQ-DiT: SliderQuant for Diffusion Models

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Abstract

Diffusion Transformers (DiTs) have recently established a new baseline for image generation models, but their major limitations are the heavy computational and memory requirements. To mitigate these bottlenecks, Post-Training Quantization (PTQ) offers a practical approach to compress these models without retraining. In this project, we adapt SliderQuant, a PTQ algorithm originally designed for Large Language Models (LLMs), to the visual generation domain. Code is available at: <https://github.com/DoctorWho28/deepLearning>.

1. Introduction

Diffusion Transformers (DiT) created a new standard of generative AI, replacing the convolutional U-Net backbone of diffusion models with a Transformer architecture. DiTs rely on **Adaptive Layer Normalization (AdaLN)** to inject continuous timestep and discrete class conditionings into each block, dynamically adapting intermediate feature states at each denoising step. DiTs achieve remarkable results in high-fidelity image generation, but their large parameter count and iterative approach lead to prohibitive memory consumption (VRAM) and slow inference speeds. To mitigate these computational bottlenecks, Post-Training Quantization (PTQ) techniques compress the weights to lower-bit formats without requiring a full model retraining.

Recently, a PTQ framework named **SliderQuant** was introduced for Large Language Models (LLMs). SliderQuant exploits the fact that different layers in deep networks have different sensitivities to quantization noise. It introduces a sliding-window distillation approach, optimizing quantization parameters over small groups of layers, and introduces learnable low-rank matrices (LoRA) to compensate for the quantization error.

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SQ-DiT aims to adapt the core concepts of SliderQuant to Diffusion Transformers. The main difference is that SliderQuant calibrates LLMs over sequential tokens, whereas DiTs require optimizing across the temporal diffusion process. During the optimization process, the weights are quantized using fake quantization and only upon completion the final weights are packed into INT8 using our custom packing implementation to effectively save VRAM space.

2. Related works

SQ-DiT derives from SliderQuant (Wang et al., 2026) adapting the quantization and distillation concepts to the DiT architecture.

PTQ4DiT (Wu et al., 2024) and Q-DiT (Chen et al., 2024) serve as our primary foundational references for the state of the art in PTQ for DiTs.

3. Quantization implementation

Because the original SliderQuant framework cannot be easily transposed, we developed the entire quantization framework from scratch.

Timestep-Aware Window Distillation. We group transformer blocks into sliding windows and optimize them sequentially. For each window, we optimize the quantization parameters by minimizing the Mean Squared Error (MSE) between the output features of the quantized blocks with their original unquantized counterparts. SliderQuant uses the L^2 norm instead of the MSE.

Unlike LLMs, where the input passes through each block only once, in DiT models the image is gradually denoised across multiple timesteps, passing through all the blocks at each step.

Since data distributions change drastically throughout the denoising process, a quantization approach that propagates error across the timesteps would be unstable. For this reason, rather than allowing the error to propagate over time, each window jointly optimizes multiple timesteps. We provide the first block with the latents pre-calculated by the

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original FP16 model across all timesteps, while the subsequent blocks receive the output of the preceding quantized block.

By eliminating temporal propagation, we align the optimization with the SliderQuant paradigm used for LLMs, that don't have noise accumulation between timesteps. This allows the LoRA matrices within each window to focus exclusively on compensating for spatial errors propagated by preceding quantized layers.

Mixed-Precision Architecture. Following the original observation that network layers exhibit varying sensitivities to quantization, we apply a mixed-precision policy. Deep and shallow transformer blocks are highly sensitive and are therefore quantized to higher bit-widths (e.g., INT8) or kept in full precision. Intermediate layers, which are more robust to quantization noise, are aggressively quantized to 4-bit (INT4). Crucial layers responsible for condition injection, such as the Adaptive Layer Normalization (`norm1`) and time/class embeddings, are entirely excluded from quantization.

Physical Packing. A common pitfall in PTQ literature is simulating quantization without realizing actual memory savings. We implement a custom physical packing mechanism (`WXAXLinear`) to collapse the learned low-rank adapters back into the main weights post-distillation. The resulting parameters are physically packed into `uint8` tensors using bitwise shift and mask operations. This guarantees that our INT4 and INT8 configurations effectively reduce the VRAM footprint of the DiT model during inference.

4. Evaluation

Table 1. Evaluation comparison

	Size	FIS ↓	IS ↑	sFID ↓	CMMD ↓
FP16	1.58 GB	24.72	318 ± 7.9	59.3	0.236
W8A8	0.97 GB	24.48	322 ± 6.0	57.8	0.239
W4A8	0.78 GB	23.49	314 ± 6.7	55.7	0.236

5. Conclusions

Statement on the use of AI

Students must include a brief statement describing whether AI tools were used in the project and, if so, how they were used. This declaration is required for transparency, but it is not intended to penalize the use of AI in any way. A project may receive any grade in the full 0 – 31 range regardless of whether AI tools were used extensively, minimally, or not at all.

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The responsibility for the submitted work always remains with the student(s). In particular, students must fully understand, verify, and be able to explain any material produced with the help of AI tools.

Misuse of AI will be penalized. Examples include submitting AI-generated material that the student does not understand, cannot justify, or cannot explain if required to do so; presenting AI-produced text, code, experimental results, or claims as if they had been independently developed or verified when this is not the case; or providing inaccurate or misleading declarations about the actual use of AI in the project.

There is no strict length limit for this section, but in most cases one column or less should be sufficient.

References

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Appendix

Adaptive Layer Normalization (AdaLN)

Sensitivity analysis

Simulation parameters